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The Impact of Technological and Institutional Innovations on Life Content: Some Time-Geographic Observations

Although they have extensively studied the diffusion of innovations, human geographers hardly have looked into the very important question of the human and societal impacts that follow the adoption of technological and institutional innovations. However, the emerging conceptual framework of time-geography lends itself to the explicit and implicit identification of far-reaching impacts upon the daily and longer-range life content of individuals. Here, some of these impacts are revealed by looking at the human individual in isolation. Other innovation impacts are specified by observing the human individual in an interdependent societal context. Several general points are illustrated with sketchy empirical evidence pertaining to the life-content impacts of the telegraph during its early U.S. diffusion.

INTRODUCTION

In highly industrialized societies, both capitalistic and communistic, the nature of technological and institutional innovations increasingly has come to influence the life content of the common individual, or the external actions and internal experiences accumulated by the ordinary person on a daily and lifelong (biographical) basis. The ways in which technological and institutional innovations express their influence upon the life content of individuals are sometimes obvious, sometimes a little obscured, sometimes rather subtle. The increased division of labor, specialization of functions and tasks, and fragmentation of human transactions required by so many modern technological and institutional

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innovations—and the pace at which such innovations chaotically appear—have contributed greatly to the disorientation, lack of coherence, and alienation sensed in one way or another by large segments of the population. The extreme role specialization dictated by technological and institutional innovations—and the consequent growing frequency with which individuals interact with one another as specialized role-holders instead of as people—has tended to make personal identification more a matter of *what* one is than of *who* one is. Moreover, since so many of the technological and institutional innovations that yield highly specialized activities require centralized administration by the physically and otherwise distant bureaucracies of ever-larger economic and governmental organizations, individuals commonly feel frustrated and powerless. And, due to their inability to affect their immediate environment, they consequently often withdraw into political inaction and other forms of apathy. In a closely related vein, the scale of activity and organization often inherent in the operation of modern technological and institutional innovations has created a widespread antagonism toward large-scale activities and institutions and the metropolitan complexes that house them, thereby inducing some to seek “simpler” environments and contributing to the quite noticeable population gains occurring in some rural and metropolitan-peripheral areas. More traditionally, it has been recognized by Marxists and others that technological innovations leading to automation in the realm of manufacturing (and elsewhere) convert workmen into “parts of a living mechanism,” putting their movements at the command of machines rather than vice-versa, and making them subject to strict *time* discipline [33, vol. 1, p. 41].

Whatever the particular ways in which a specific technological or institutional innovation may affect the life content of a person, whatever the mental experiences and reactions it generates, it will always carry a two-fold general impact. First, it will require interaction with other humans, man-made objects, or elements of the natural environment. Second, it will make claims upon that most inflexible limited resource of an individual—real sidereal, or measured, time. This two-fold demand, in turn, means that the impact of technological and institutional innovations upon individual life content is usually far reaching and profound in ways that are apt to be totally missed by those who assess innovations solely in economic or outwardly ideological terms of one kind or another.

Since time is a scarce resource, and not merely a medium through which events pass or a passive backdrop before which events unfold, its use in conjunction with the interactions prescribed by an innovation affects both the amount remaining to an individual and his freedom to participate in other activities, or interactions, with other people, artifacts, and elements of the natural environment. More precisely, a great deal of the daily and biographical impact of any technological or institutional innovation upon individuals results from the intersection of pervasive *time-geographic realities* with certain properties of the innovation itself, the most important of which are the sector of life it directly affects (e.g., work, education, medical care, religion) and *its specific temporal and spatial attributes*.

Whether innovations are dramatic, or seemingly inconsequential, the most important time-geographic realities contributing to their impact on the life

content of human individuals are the following:¹

1. Every human being is physically indivisible. He cannot be at more than one physical location at any point in time. Or, in tracing out his path, or trajectory, of existence through time-space at daily and lifelong scales, he cannot participate simultaneously in activities that are spatially separated from one another. (All inanimate objects, such as machines, tools, and buildings, and all nonhuman living beings, such as animals and plants, are also indivisible “individuals” subject to the same simultaneity constraints.)
2. Every task or activity that a human individual conceivably can undertake is time demanding. Hence, every commitment to a task or activity diminishes both the finite daily (or lifelong) time resources of the participating individual and the finite daily time resources of the population as a whole within any bounded area. (The total daily time resources of the “population system” of any bounded area are equal, obviously, to 24 hours multiplied by the number of persons inhabiting the area.) Put alternatively, whether viewing one human or an areally aggregated segment of “society” as a whole, only so many tasks and activities may be packed into a single day.
3. Movement between any pair of points in space can only be accomplished through the sacrifice of time. Therefore, a person can only take part in two successive activities, *x* and *y*, if they are so located in space that the distance separating them can be traversed, *with the transportation mode at his command*, in the time interval occurring between the termination of *x* and the commencement of *y*.
4. Space has a limited packing capacity, or ability to accommodate events, no two physical objects being able to occupy the same space at the same time. Thus, any set of activities whose occurrence requires the use of uniquely designed physically fixed structures or objects must be placed spatially apart from one another, *regardless of their timing*. (Of course, this does not preclude two such activities from being physically adjacent to one another.)
5. Insofar as a given “activity bundle” definitionally consists of a specific set of roles, the bundle becomes closed to additional participants once those roles are fully staffed. (“Activity bundles” are formed when the paths of two or more individuals converge in time at a particular location [“station” or “domain”]).
6. Because realities 1 to 5 are always true, regardless of individual and cultural variations in the perception, conception, and definition of time,²

¹Length restrictions make it impractical to fully depict the principal features of the still evolving time-geography framework. Moreover, although much of the pertinent literature remains in Swedish, numerous English language publications contain either summaries of the framework as a whole, or an elaboration of specific time-geographic concepts [11, 14, 16, 19, 20, 23, 28, 30, 37, 41].

²In other words, even when time perceptions, conceptions, and definitions are not sidereally based, or even when “real” measured time is distorted or totally ignored for psychological or cultural reasons, time-geographic realities are operative, impacting upon life content by affecting the ability of individuals to participate in different activity sequences. For a discussion of different psychological, cultural, academic, and historically based conceptualizations of time see [9, 45, 47].

there is always a complex *local connectedness* between the apparently unrelated and unconnected events and activities experienced by different individuals. That is, collateral processes, each of which has its own time-geography, constantly impinge upon one another. (The more academic specialists assign events and activities to arbitrarily defined, context-divorced categories of analytical convenience, the more “invisible” the local connectedness of phenomena tends to become [14, 15, 22].)

Using the just-enumerated inescapable time-geographic realities as a point of departure, this paper attempts to integrate and considerably extend ideas previously presented on the time-geographic consequences of innovations [6, 7, 8, 18, 37], illustrating specific points wherever possible with skeletal empirical evidence pertaining to the life-content impacts of the electromagnetic telegraph during the early (post-1844) period of its diffusion in the United States. The telegraph is especially appropriate for illustrative purposes, partly because in sweeping its pageant of change across the country it directly and indirectly touched upon the life content of so many common individuals, and partly because it was the first innovation greatly to facilitate the long-distance administration and control of business and government units from centralized locations, thereby serving as a forerunner to many present-day technological and institutional innovations.³

LIFE-CONTENT IMPACTS: THE HUMAN INDIVIDUAL SEEN IN ISOLATION

Life-Path, or Biographical, Impacts

Almost every modern technological and institutional innovation applied outside the household results either in the direct creation of new specialized roles that have *an independent existence of their own*, or in the modification or redefinition of such roles. The roles fostered or altered by such technological and institutional innovations are independently existing (until terminated or superceded) in the sense that when they are not filled by one person they sooner or later must be filled by another person (Fig. 1). Or, the people whose life paths pass in and out of the detailed roles that become embodied in the economic organizations and other institutions of highly industrialized capitalistic and communistic societies are impersonally interchangeable so long as their earlier life paths have equipped them with the skills or competencies required by those roles. (The specialized roles—lawn-mower user, dishwasher user, and so forth—associated with the application of modern technological innovations within the household are less impersonal insofar as they, like many preindustrial economic roles, often can be easily delegated on a temporary and spontaneous basis from one role-holding household member to another.)

Quite plainly, the broad biographical features and experiences of the individual whose life path has linked up with an innovation-induced role for weeks,

³The sketchy empirical evidence on the telegraph presented here is developed at much greater length in a forthcoming book.

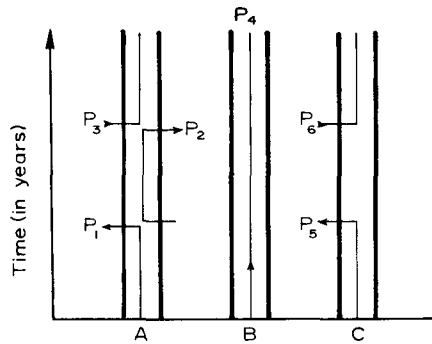


FIG. 1. Independently Existing Specialized Roles Resulting from a Technological or Institutional Innovation. Over any prolonged period of observation some roles independently existing (e.g., A) may be filled by a succession of people without becoming vacant for any extensive time. Other independently existing roles may not change hands (e.g., B). Yet others may become temporarily vacant due to the difficulties of finding a qualified replacement (e.g., C). P_1 – P_6 represent separate role-holding people.

months, or years are by definition different from those he would have acquired if the newly created or defined roles had not materialized, if he had not found his way to that particular role vacancy, and if large portions of the same weeks, months, or years had been otherwise allocated. For example, the people who became telegraph operators, message copiers, messenger boys, clerks, battery tenders, line guardsmen and repairers, local office superintendents and managers, or company-wide office secretaries, auditors, and administrators during the late 1840s and 1850s⁴ [26, 43, 46] (when commercial use of the telegraph spread from Washington and Baltimore to hundreds of other U.S. urban centers), would have existed all the same if the telegraph never had diffused as an innovation and never brought their particular specialized roles into existence. But they would have been forced to find other employment roles within their respective local “activity systems,” or to fill their lives with nonpaying tasks while going unemployed, or to migrate to another locality.

As a consequence of being locked into a new or modified role associated with a technological or institutional innovation, a person may have the length of his biography, or life path, reduced by conditions associated with the role; e.g., exposure to noxious chemicals, high noise levels, extreme stress, or extremely debilitating interference with natural biological rhythms. Also, if that role is an employment role, a person’s ability to incorporate certain other activities and experiences into his life content will depend in some measure upon the income it provides. In addition, under some circumstances the holder of an innovation-derived role eventually may find that the place in which his life content is

⁴Ironically, not only did the telegraph make centralized administration and control more feasible in general, but the large telegraph companies that emerged from the merger activities of the 1850s were themselves among the very first large economic organizations to control geographically widespread units from a centralized office.

accumulating must be shifted, either because the locus of the role has been changed, or because the role-controlling organization or institution suggests or demands that a new and differently located role be assumed. Thus, many operators and other individuals employed by telegraph companies during the mid-nineteenth century found their overall life content drastically altered as their life paths were steered from office to office. One person, for example, was shifted about from Wilmington (Del.), to New Hope (Pa.), to Newark (N.J.), to Philadelphia, to New York, and back again to Philadelphia [43].

Although modern technological and institutional innovations almost always lead to the creation or reshaping of independently existing specialized roles, they often simultaneously eliminate other previously existing specialized roles. This is the case, for instance, when industrial or transportation innovations diminish the number of specialized employment roles provided by older, less efficient means of production or movement; or when administrative innovations lead to the substitution of one set of procedures, functions, and specialized roles for another. Obviously, whenever specialized employment roles are eliminated, the way in which the life content of any single previous holder piles up is changed dramatically, in that his life path must be channelled into another employment role, perhaps one attached to the activity system of another place or area, or he must be content with unemployment and whatever interactions and experiences it permits.

Daily-Path Impacts

The use or consumption of virtually all technological and institutional innovations, whether adopted and applied outside or inside the household, necessitates at the very least that one human individual spend some part of a day, which may or may not be temporally fixed by set operating or availability hours, at one or more specific locations, or "stations." Unless the innovation is a time-saving substitution for an already used innovation (e.g., a task-shortening piece of machinery or a more rapid mode of transportation), this will require some time-use and activity displacement. Or, the person's daily path will become infused with new content, requiring that he either sacrifice, reduce in duration, or perhaps reschedule some previous participation in a production, consumption, social, or leisure-time activity [31].⁵ If use or consumption of the innovation is only possible at a "station" at which the user would not normally find himself, then an additional amount of time would also have to be spent moving to and from that station. This, too, would influence the person's daily life content insofar as it demanded the total or partial abandonment of yet other time-consuming, space-specific activities.

Just how many adjustments to daily life content are associated with the new use or consumption of a given non-time-saving innovation depends on several

⁵With the use or consumption of some technological or institutional innovations individual time-use and activity displacement may be seen as occurring on weekly, seasonal, or other scales exceeding that of the day.

factors of varying obviousness:

1. *The minimum time required for use or consumption.* The more time is required for use or consumption, the less daily time will be available for other activities, regardless of their temporal and spatial characteristics.
2. *Whether or not innovation use or consumption is temporally fixed, or confined to specific hours, and whether or not activities of previous involvement are temporally fixed.* Because the human individual is indivisible, any temporally fixed activity that partly or fully overlaps with a new temporally fixed activity must be partly or totally forfeited. In fact, if forfeiture turns out to be too great a sacrifice, or conflicts too greatly with the person's established daily pattern of existence, the innovation eventually may have to be rejected. (This is admirably illustrated by Carlstein's analysis [6] of the impact of elementary school operation upon the daily paths of children in the Mexican village of Tepoztlán.)
3. *The spatial location of the station at which innovation use or consumption must occur vis-à-vis the spatial location of other activities—whether temporally fixed or flexible—that previously have been included in the daily path.* The greater the physical distance separating the innovation-use station from previously used stations, the greater the activity displacements that must occur.
4. *The prevailing speeds and schedules of the mode(s) of transportation available to the person.* The faster the physical distances to and from the innovation-use station can be covered, the fewer the activity displacements that otherwise would be necessary (Figs. 2–3).

Some of the most extreme modifications of daily life content occur when innovation use involves nighttime employment and the postponement of physiologically mandatory sleep to hours that coincide with the operating hours of many, if not most, of the activity stations at which the person previously stopped. By severely conflicting with a person's natural biological rhythms, innovations requiring nighttime work also greatly influence internal bodily and psychological experiences, and thereby daily life content.

When a modern technological or institutional innovation is a time-saving substitution for an earlier used innovation, it definitionally releases time for interactivity travel and additional daily activity-participation or social-interaction combinations. Precisely what additions a person can make to his habitual daily path will depend on exactly how much time is saved, the fixed or flexible hours of both existing activities and additional activity or interaction alternatives, the spatial location of those existing activities and alternatives, and the speed(s) at which he can move, or whether or not the alternatives fall entirely within the time-space prism, or area, within his reach (Fig. 4). Moreover, because of time-geographic realities, if the total time savings to a person from a number of technological innovations are not available in one or two solid blocks, but chopped up in numerous small interactivity pieces, the possibilities of his adding new activity stations to his daily repertoire are rather limited [33]. Therefore, it might be argued that increases in so-called leisure time subject

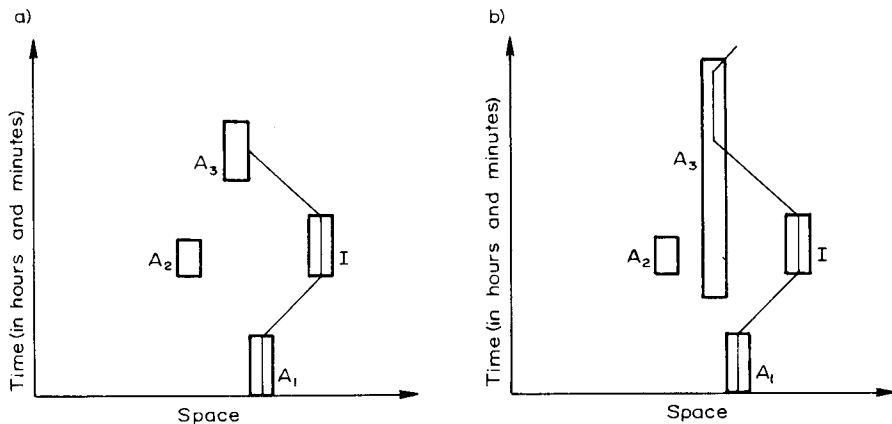


FIG. 2. Hypothetical Daily Path Impacts of Temporally Fixed Innovation Use or Consumption. *a.* Activities A_1 – A_3 were all previously elements of the observed person's partial daily path. Each of these activities is temporally fixed and requires that all participants arrive no later than its starting time and remain throughout its full duration. The new innovation use or consumption activity I also shares these characteristics. Participation in I requires that A_2 be displaced because it occurs simultaneously with I . A_3 must also be displaced because the person would arrive too late. Note, however, that A_3 would still be possible either if it had another location physically closer to I , or if the person in question had a faster means of transportation available to him (i.e., a less steeply inclined interactivity path). *b.* The conditions prevailing are identical with those in case *a*, except that A_3 is temporally flexible, allowing a person to start at a variety of times but requiring the same duration of participation as previously. Innovation use or consumption at I still necessitates the sacrifice of A_2 , but no longer displaces A_3 . Note, however, that A_3 would have to be displaced either if it were somewhat more physically distant from I , or if the person in question had a slower means of transportation available to him (i.e., a more steeply inclined interactivity path).

individuals to greater clock tyranny, since they “now have to time their [growing number of] activities more and more precisely” [29, 47].

During the early years of its operation, the telegraph affected the daily paths, and thereby the daily life content, of both those who worked for the telegraph companies and those who went to telegraph offices to send messages. For operators and others who were employed in mid-nineteenth century telegraph offices, the working day was extremely long, even by then prevailing standards. Except on Sundays, the workday customarily ran from sunrise (or earlier if important news was to be forwarded) until 10 P.M. (or later, since offices were not to close if unsent messages and news dispatches remained), with an average length of sixteen hours not being unusual [26, 32, 43, 46]. (In contrast, the workday of the factory hand in the 1850s apparently was more generally in the range of eleven to twelve hours [25].) This meant that there was precious little opportunity for the telegraph-office worker to include nonwork interactional activities in his daily path, partly because most of his “free” hours had to be allocated to sleep and other physiological necessities, and partly because the operating hours of few, if any, other activity stations coincided with his daily work-free periods. Moreover, if the employee was associated with an office in

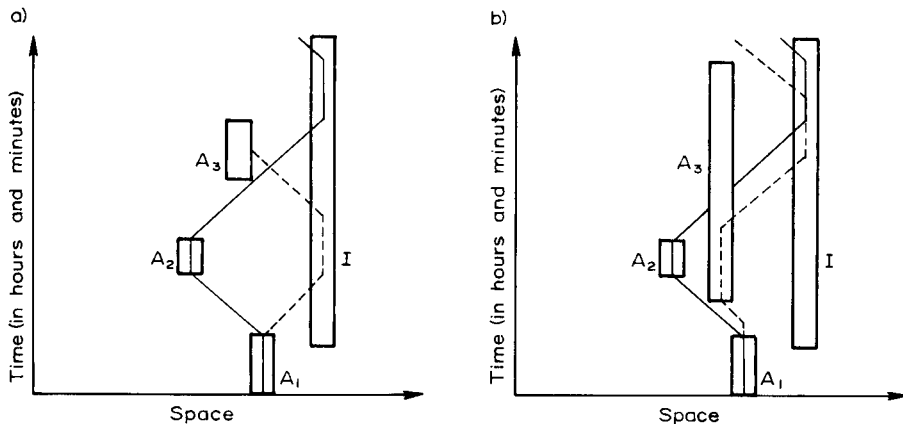


FIG. 3. Hypothetical Daily Path Impacts of Temporally Flexible Innovation Use or Consumption. *a.* The conditions prevailing are identical with those obtaining in Fig. 2*a* except that *I* is temporally flexible (although still demanding the same length of participation). In contrast to the situation in Fig. 2*a*, it is no longer necessary to displace both *A*₂ and *A*₃ as a result of innovation use or consumption at *I*. *A*₂ can be retained (solid daily path). But *A*₃ must still be forfeited because of late arrival, even if *A*₂ is dropped from the daily path (dashed daily path). Note, however, that *A*₂ would also have to be omitted, despite the greater flexibility of *I*, if the person in question had to use a slower means of transportation. *b.* The conditions prevailing are identical with those in Fig. 2*b*, except that *I* is again temporally flexible, while still requiring the same amount of participation time. Now either *A*₂ (solid daily path) or *A*₃ (dashed daily path) may be saved from the pre-innovation-use daily path, although the retention of both remains impossible due to "coupling constraints." Even the availability of a faster means of transportation would not allow the person in question to preserve *A*₂ and *A*₃ after committing himself to *I*.

the heart of the business district of a large city, the possibilities for enriching his daily life content with other activities were even further diminished by the relatively long journey to work that he was likely to take by foot. (The high fares of horse-drawn trolleys usually put their regular daily use beyond the reach of those with normal incomes.) Although telegraph messenger boys had much less spatially confined daily paths, and thereby much more varied types of experiences during working hours, their chances for participation in nonwork activities were no less constrained than those of their fellow workers, especially since their very low incomes definitionally would have prevented them from gaining access to many types of stations and domains even if they had the time.

Highly irregular working hours characterized the plight of those hired to repair and maintain telegraph lines during the late 1840s and 1850s. In some measure owing to an as yet unperfected technology, in some measure owing to the cheap and careless construction techniques employed by profit-greedy contractors, the functioning of almost all early telegraph lines was frequently unreliable and ephemeral. Wires were often grounded by heavy winds or sleet, poles often collapsed after rotting if they had not been earlier blown down or washed away, poorly jointed wires interfered with transmission, and faulty insulators were constantly troublesome. The persons employed to cope with

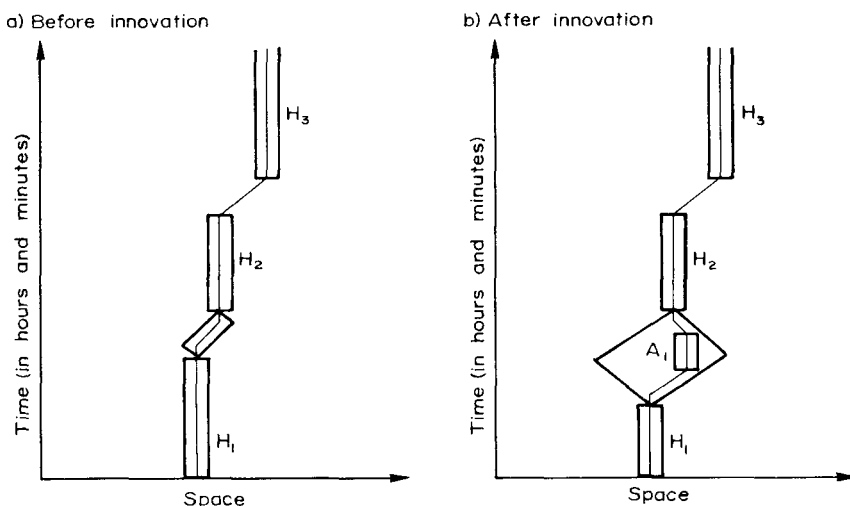


FIG. 4. Hypothetical Daily Path Impacts of a Time-Saving Innovation. Activities H_1 – H_3 are habitual elements in the weekday daily path of a person. A time-space prism separating the ending time of H_1 and the starting time of H_2 is indicated in both diagrams. The outer turning points of these prisms are determined by the speed of the means of transportation available to the person. It is impossible for him to appear outside the lower walls of the prisms because they are physically inaccessible by definition. It is also impossible for him to appear outside the upper walls of the prisms because he then would not be able to reach H_2 before its starting time. Before an innovation reduces the time that must be allocated to H_1 , there is little in the way of activity or interaction that is feasible for the person prior to the start of H_2 . Consequently, he takes his time leaving H_1 and arrives at H_2 somewhat early. After a time-saving innovation allows H_1 to be completed more rapidly, a much larger time-space prism becomes available to the person and, consequently, other tasks, errands, or social interactions, such as A_1 , can be incorporated into his daily path.

these problems often found themselves searching at night for breaks by foot or train, or getting up at 4:30 A.M. to test for current, or patrolling the line—sometimes as much as twenty to fifty miles from base—as a preventive measure [26, 43, 46]. Under these circumstances the person who became a repairman or maintenance man for a telegraph company had to occasionally or permanently relinquish all previous activities that required presence at a fixed temporal location. Now and then he also had to displace more temporally flexible activities when he was either too distant from the station involved, or forced to sleep during daylight hours.

Since charges were high, most telegraph message senders, other than newspaper employees, were businessmen. At first, when there was only one office in a city (or a number of very closely situated competing offices), and when the infant technology required that messages to specific places be sent within limited hours, the daily paths of these consumers quite likely had to be rearranged, and some customary activities had to be partially or totally displaced. Activity displacement in New York, Philadelphia, Boston, and other large cities was apt to occur due to: queuing delays during the hours of heaviest

telegraph use (1 to 3 P.M.); and simultaneity conflicts either with closely spaced business meetings or the operating hours of the local merchants' exchange (which ceased business sometime between 2 and 4 P.M., depending on the city), U.S. customhouses (which shut up at 3 P.M.), and banks (which closed their doors at 2 or 3 P.M.) [2, 25].⁶ Later on, when numerous auxiliary telegraph offices were opened at hotels, railroad stations, and other large-city locations, it must have been easier for many individual message-sending businessmen to maintain the most important elements of their usual daily paths or even add new activity components to those paths. This would have been the case insofar as less time than previously had to be allocated to reaching the telegraph office, and insofar as telegrams were substituted for letters and the branch telegraph office used was closer than the city's lone post office. (Until 1858, no U.S. city, including New York, had more than one post office, and even then mails for given locations had to be deposited by specified hours if they were to leave the same day [25, 35].) Also, when telegraphic technology eventually permitted messages to be sent to any location at any office hour, the chances for simultaneity conflicts were considerably reduced. Finally, when intraurban telegraph lines were established for local commercial purposes, businessmen could eliminate travel time and free up time for additional activity participation and social interaction whenever they substituted telegraphic exchanges for face-to-face meetings.

LIFE-CONTENT IMPACTS: THE HUMAN INDIVIDUAL SEEN IN HIS INTERDEPENDENT SOCIETAL CONTEXT

Because, among other reasons, there is an intricate local connectedness of events and activities, the impact of technological and institutional innovations on the life content of a particular individual seldom, if ever, occurs in a vacuum. Societal interdependencies are such that the use or consumption of an innovation by one person almost invariably touches upon the activity participation and life content of one or more other persons. This may happen in a variety of ways, some of the most significant of which are spelled out below.

Population System-Activity System Matching Dynamics

The "activity system" of any given bounded area (or all those time-packaged activities actually carried out locally *plus* any independently existing roles that are temporarily vacant) and its corresponding "population system" (or all the

⁶Regardless of the historical setting, the more one can specify about the operating hours, possible queuing times, and spatial coordinates of business, social, and public sector activities, and the more one can specify about public transportation routes and schedules and other travel times, the more the time-geographic methodology will permit one to say something about what daily paths are or are not possible for a representative hypothetical individual once he has added a new activity to his daily program.

area's inhabitants and household groupings) are constantly engaged in a matching process [11, 21, 41]. Put abstractly, each of the time-packaged roles and activity bundles contained within the activity system seeks a bearer (or bearers) among the population while, simultaneously, members of the population system pick their way to a series of roles and activity bundles while seeking outlets for their needs, wishes, and competencies and allocating their finite time resources.⁷ Technological and institutional innovations contribute to the dynamics of this matching process insofar as they lead to the creation or elimination of *sets* of self-existing specialized roles and activity bundles (each of which affect the life paths of more than one person at a time).

When a person moves into an innovation-spawned role, he may touch upon the life content of others by blocking the role off from other aspirants or by abandoning a previously held specialized role, thereby creating an opening for another person whose life would have been otherwise shaped. Furthermore, the appearance of a set of independently existing employment roles in conjunction with a technological or institutional innovation is quite apt to set complex "chains of opportunity" [51] in motion whereby the roles are filled by already locally employed persons who switch jobs, only to have their newly vacated roles filled by other already locally employed persons, who in turn have their newly surrendered roles filled by yet other employed persons, and so on until population-system members who are not currently employed are able to assume abandoned roles (Fig. 5). Moreover, if new specialized roles cannot be filled directly, or if chains of opportunity cannot be broken by local persons with the requisite time resources or competencies, then the in-migration of persons from other areas is likely to occur—a situation that definitionally will have ramifications for the population system-activity system matching dynamics of those areas.

When a person is pushed out of an independently existing specialized role as a consequence of that role's being terminated by a technological or institutional innovation, he may influence the life content of others by redirecting his life path into another local role and blocking that role off from other aspirants. Or, in response to the circumstances, he may also migrate out of the area, thereby changing the lives of those household members, relatives, and friends he either takes with him or leaves behind. And, if migrating to another place, he there will cause otherwise absent population system-activity system matching details, or a shuffling of the life content of other people as a consequence of his participation in various activity bundles and social interactions.

Innovation-Use and -Consumption Projects

In most instances the use or consumption of a technological or institutional innovation cannot be fully accomplished through the time allocation of only one individual at a specific location. Instead, the use or consumption of such innovations usually requires that the time-space trajectories of several people be

⁷Cf. time-geographic reality number two.

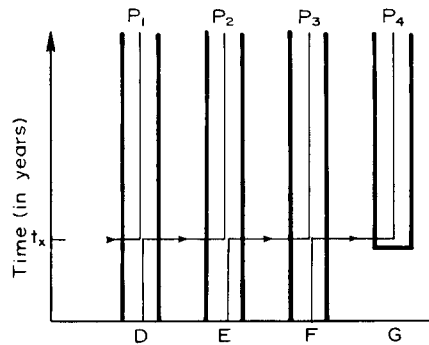


FIG. 5. A Local Employment-Role "Chain of Opportunity" Resulting from a Technological or Institutional Innovation. Column G represents a specialized employment role, or job, created at date t_x in conjunction with a technological or institutional innovation. Columns D, E, and F represent specialized employment roles brought into being by previous innovations. P_1 – P_4 are persons who are members of the local population system.

bound up with physically tangible resource inputs of one kind or another (e.g., machinery, raw materials, buildings), or with the life paths of nonhuman and inanimate "individuals." (Physically intangible information or energy is also generally necessary to innovation use or consumption.) Very frequently, all the "individuals" participating in the use or consumption of an innovation are not brought together at a single time-space niche. They are rather joined in an orderly fashioned "*project*," or a specific sequence of differently combined activity bundles occurring at different moments and spatial locations; as, for example, in the case of manufacturing and other production processes, or in the provision of health, educational, and cultural services [18, 48].

By definition, through its synchronization and "synchorization" demands, each activity-bundle phase of an innovation-use (or -consumption) project preempts: land or building space that, for the duration of the phase, cannot be used by outside people for other purposes (rules, laws, and power relationships, or "authority constraints," may further prevent use of the space by others during times of the day or year when the project phase is not in operation); resource inputs that cannot be exploited by other potential users;⁸ plus the interactivity travel time and simultaneous activity-participation time of numerous human individuals. Recalling the daily activity-displacement consequences that arise even when only one innovation user or consumer is seen in isolation, it should be evident that the constraint impacts of innovation-use projects on the formation of other activity bundles may spread extremely far

⁸In keeping with the idea that inanimate objects trace life paths, it ought be kept in mind that resources, unless naturally present, do not simply materialize for utilization in a project activity bundle. "What actually is available is completely determined by how transportation and storage [and their related economic factors] have come to distribute potential [resource] inputs beforehand up to the critical moment when a need is felt" [18] within the time-space volume occupied by the activity bundle.

both temporally and spatially. Just what the collective magnitude of activity-displacement effects following from any innovation-use project will be depends, among other things, largely upon the number of separate activity bundles required by the project, the scale and duration of each of those bundles, and the temporal and spatial proximity of successive bundles to one another.

The ability of an innovation-use (or -consumption) project to touch upon the life content of numerous people through participation (and thereby also activity displacement or rescheduling), and to preempt resource inputs (however meager) can be illustrated by looking upon something as relatively simple and unremarkable as the completion of a telegraph-message circuit under mid-nineteenth century conditions as an innovation-consumption project and dissecting its component activity bundles (Fig. 6).

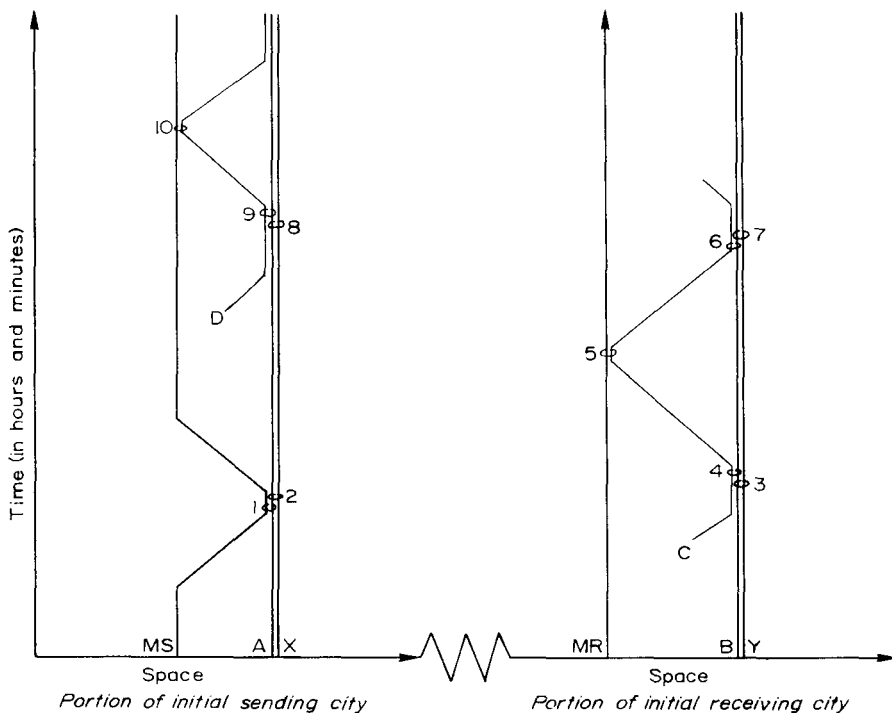


FIG. 6. The Completion of a Telegraph-Message Circuit under Mid-Nineteenth Century Conditions as an Innovation-Consumption Project. See text for letter designations and elaborations. 1-10 represent the successive activity bundles contained within the project. *Project completion requires that these bundles be sequenced in no order but this.* For the sake of clarity there is no indication of the paths of telegraphs, pieces of paper, and other project resource inputs, or participating inanimate "individuals." For the same reason, the paths of C and D are not shown for the entire duration of the project, and other (non-project) activity bundles and interactions involving the project's human participants are not depicted.

Activity bundle one—message deposit: The daily paths of a message sender (MS), or innovation consumer, and clerk A are united at a telegraph-office counter, possibly only after the sender has allocated some time to queuing up behind other senders. In writing out his message the sender preempts the use of a pen and a piece of paper (or intersects with their life paths).

Activity bundle two—message transmission: Clerk A has now spent a small amount of time moving to another part of the office where his daily path joins that of telegraph operator X in order to give him the message. In transmitting the message, operator X preempts the use of a telegraph and the table to which it is attached, as well as the paper upon which the message is written.

Activity bundle three—message arrival at another office: Operator Y in another city, using another telegraph and table plus pen and paper, hears the message and transcribes it before handing it over to clerk B, whose daily path has momentarily linked up with his.

Activity bundle four—arrangement for local delivery: After the clerk in the receiving city devotes a little time to moving to another part of the office, his path temporarily coincides with that of a messenger boy (C) who is given the message, or paper, and delivery instructions.

Activity bundle five—local delivery: Having consumed some time while walking or riding through city streets, the messenger boy brings his daily path into conjunction with that of the intended receiver (MR) and hands him the paper containing the message.

Activity bundle six—response deposit: The daily path of the intended receiver or the messenger boy, now acting as sender, meets up with the daily path of clerk B and, with another pen, a response is committed to paper.

Activity bundles seven through ten—response transmission, response arrival at the initial sending office, arrangement for local delivery, and local delivery: these four bundles resemble bundles two through five, except that: the roles of clerk A and telegraph operator X are reversed with those of clerk B and operator Y; another messenger boy (D) joins the project to follow a delivery route in the city from which the original message was sent; the initial sender becomes the intended receiver; and different times are involved.

In comparison to what has just been described, the sequencing and individual details of activity bundles associated with the use projects of many present-day technological and institutional innovations are far more intricate and demanding of temporal exactitude [45, 47] and coordinated synchronization and synchronization. In fact, the fixed spatial attributes and temporal rigidities of the innovation-based production and consumption projects embedded in the contemporary activity systems of *all* highly industrialized countries are such that it may be argued that the needs and wants of the population system, or people, have come to respond and adjust in increasingly great measure to the “needs” and “wants” of the activity system, or technologies and institutions and their independently existing roles. (Those ignorant of time-geography, among others, might assume the opposite to be more true.)

The project-based claims made by technological and institutional innovations upon the life content of interdependent human individuals are compounded when the inputs of one innovation-use project represent the outputs of one or

more other innovation-use projects or sub-projects, as is characteristically the case in modern extended production systems where the projects and sub-projects of a great many geographically separated highly specialized plants and administrative units lock into one another [38, 48]. More precisely, under these circumstances each iteration of an innovation-use project or sub-project may require activity-bundle formation both prior to its start and subsequent to its finish, including the coupling together of individual human paths so as to facilitate contracting and subcontracting agreements, orders, deliveries, the clearing up of technical problems, and other *intraorganizational* and *inter-organizational* matters.

The fact that different projects lock into one another, or are backwardly and forwardly linked, means that changes in the scale of one innovation-use project can generate local or nonlocal multiplier effects in other innovation-use projects. Since these multiplier effects are synonymous with increases or decreases in the number of independently existing roles associated with other innovation-use projects, they have ramifications for population system-activity system matching processes occurring both locally and nonlocally. Hence, for example, chains of opportunity involving the life content of many persons can be triggered locally and nonlocally, and in-migration or out-migration may touch upon the life content of persons from the local area or other places (Fig. 7). The number of life paths affected by innovation-use project scale changes may be further increased by: (1) the income expenditure multiplier effects, or specialized role increases, created by project role holders; and (2) second- and

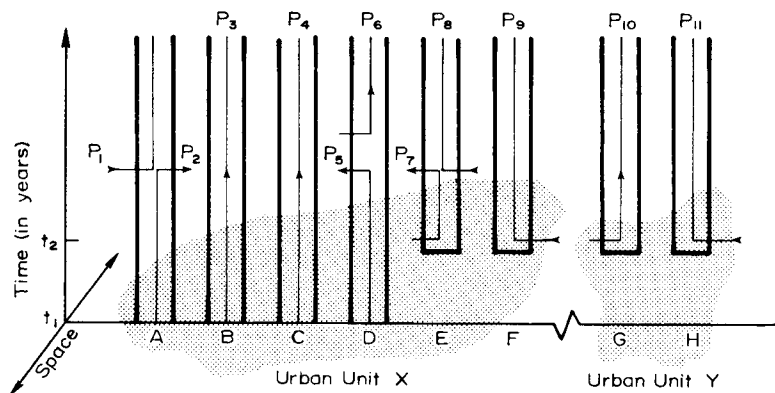


FIG. 7. Specialized Employment Roles, Multiplier Effects, and Migration Resulting from a Technological or Institutional Innovation-Use Project. Specialized employment roles, or jobs, A-D represent a sample from the total set of independently existing roles directly resulting from the expansion or inception of an innovation-use project at urban unit X at date t_1 . By date t_2 the technological or institutional innovation-use project generates both local employment multipliers (represented by employment-role sample E-F) and nonlocal job multipliers at urban unit Y (represented by employment-role sample G-H). P_2 - P_8 are "native" members of urban complex X's population system, and P_1 and P_9 are in-migrants to X. P_{10} is a "native" member of urban area Y's population system, and P_{11} is an in-migrant to Y.

later-round multiplier effects originating in those projects backwardly and forwardly linked to the project that initially altered its scale.

*Projects Involving the Pre-Use Construction of
Innovations and Their Facilities*

Before certain technological or institutional innovations (such as factories, forms of communication, and new service types) can be used or consumed, it is first necessary either that specially designed buildings or other facilities be constructed, or that the innovation itself be assembled or produced. These pre-use construction and related activities in themselves constitute “projects” that have time-geographic attributes and impacts on the life content of those participating in constituent activity bundles. The capacity of such a project to enter into the life content of a great number of people by necessitating activity-bundle participation—and displacement—can be exemplified by briefly describing the activity-bundle formations subsumed by the mid-nineteenth century construction and establishment of telegraphic facilities between U.S. cities.

The construction and establishment of telegraph lines between places typically required at least three distinct sub-projects whose durations frequently overlapped with one another [1, 3, 12, 26, 32, 42, 43, 44, 46, 49].

First of all, the telegraph company intending to operate a line would have to complete a number of preliminary arrangements.⁹ This sub-project included: the procurement of right-of-way privileges from private landholders, public road authorities, or railroad companies; the negotiation of a contract with a line-construction contractor; and the acquisition of capital sufficient to cover the costs of construction, which, depending on topography and other conditions, normally ranged from \$100 to \$250 per mile for a single wire plus \$100 per mile for each additional wire that proved necessary. Each of these arrangements required a more or less logical sequence of activity bundles to be formed in which the paths of company representatives were joined up repeatedly with prospective right-of-way leasors, competing contractors, and bankers and other potential capital suppliers. The number of interactions necessitated by capital acquisition were apt to be most numerous, occurring simultaneously in several cities and often transpiring after the second sub-project was under way.

The second sub-project consisted of the actual construction of the telegraph line (Fig. 8). During each day of construction the paths of hole diggers, pole setters, wire stringers, foremen, and work-party cooks were joined in groups of five to fifty to form various activity bundles along the line’s intended route. (Construction sometimes took place simultaneously along different route segments, and hole diggers occasionally formed separate bundles working in advance of everybody else.) Since the people involved in the sub-project usually could not return home at night, instead sleeping in mobile tents at the advancing edge of construction, all those activities that required their presence

⁹The formation of the telegraph company itself also can be depicted as a project or sub-project.

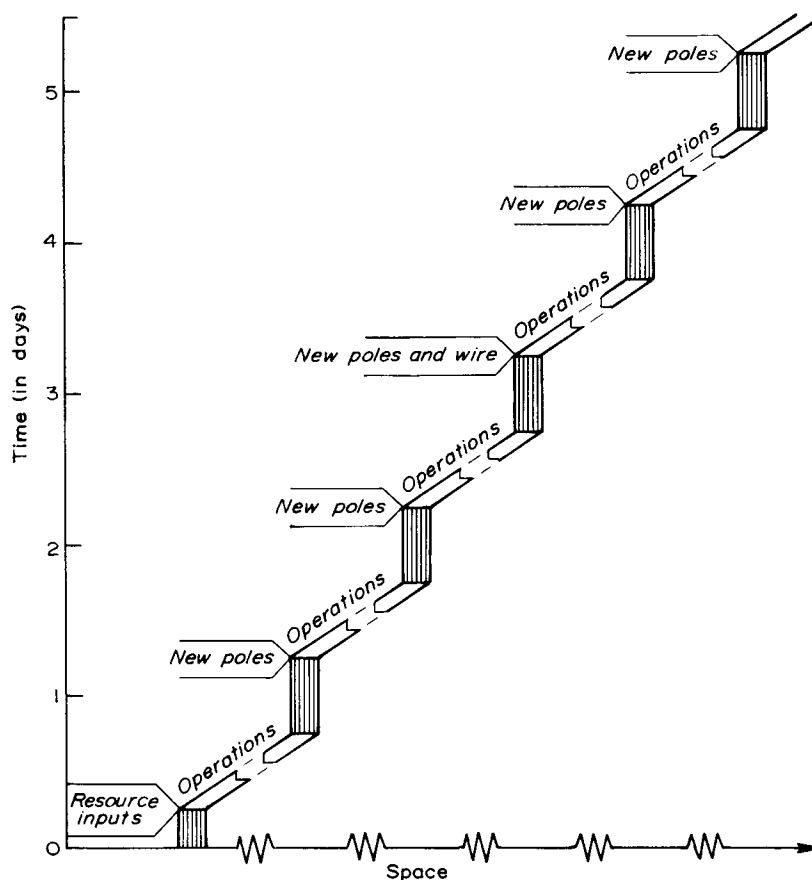


FIG. 8. Mid-Nineteenth Century Telegraph Line Construction as an Innovation Pre-Use Sub-Project. Vertical lines represent persons holding sub-project specialized roles (see text) while encamped for the night at the advancing edge of construction. Within the time-space area, or "pocket of operations" [18], of each day's construction, activity bundles are formed at every pole site. There a hole is dug, the pole is set, and the wire(s) and insulator(s) are attached to crossbars. Since bundles could be formed at 125 to 300 or more sites per day, paths are not indicated within any daily "pocket of operation." See text for initial resource inputs of the sub-project.

at stations located elsewhere had to be displaced. The day-to-day operations of this sub-project were very sensitive to the synchronized and synchored appearance of resource inputs, normally including: iron wire (varying with quality from 250 to 400 pounds per single-wire mile); gutta-percha-encased cables for crossing rivers and streams; regularly spaced poles (25 to 40 per mile) that were cut without payment from neighboring forests, or cheaply purchased from nearby farms, or prepared and shipped in from great distances when trees were locally absent; wooden crossbars for attachment to the poles; nuts and bolts for securing the crossbars; insulators to be set in the cross-bars; ground

augers and miscellaneous tools; tents and blankets for sleeping purposes; cooking equipment, provisions, and utensils for eating purposes; plus wagons, and horses, mules, or oxen to draw things over the five to eight miles generally traversed in the course of a single working day.

In order for these resource inputs to arrive at the right places and right times, the daily paths of the supplying farmers, warehousemen, and others had to be swept temporarily into the sub-project. And, when synchronization and synchronization problems arose—whether due to the weather, input shortages, or a lack of funds to pay suppliers—the resulting delays caused further activity displacements for those people working directly in the sub-project.

The final sub-project involved the obtaining of office space (new offices seldom were constructed expressly for the purpose) in each of the towns and cities through which the telegraph was to pass, as well as the outfitting of those offices with telegraphs, the batteries necessary to run them, furniture, and paper and pens. The carrying out of this sub-project entailed the formation of orderly sequenced activity bundles wherein the paths of company agents, or their contractors, were joined with those of potential leasers, telegraph and battery manufacturers, office equipment retailers or wholesalers, and shippers.¹⁰

It is to be further realized that the life-content impacts of the latter two sub-projects were extended to yet other persons insofar as they instigated the creation of additional self-existing specialized roles, or multiplier effects, at places such as New York, Pittsburgh, Cleveland, and St. Louis where iron wire, telegraphs, insulators, and other consumed items were manufactured.

Household Interactions and Activity Bundles

It is quite clear that a number of things that are at the very core of a person's life content—his childhood socialization, his personality development, his formation of attitudes, his sexual identity, his capacity for emotionally dealing with others, and so forth—are much rooted in the accumulation of household, or family, interactions¹¹ that he corporeally participates in and internally experiences as both a child and an adult. However, especially under modern circumstances, just what household activity bundles one does or does not participate in, under what conditions, and when, *is in considerable measure determined by the time-geographic attributes of the technological and institutional innovations used by different household members while away from their home or apartment* [13, 17, 30, 31, 36].

The burden of activity displacement and rescheduling phenomena discussed in connection with both the daily paths of observationally isolated human individuals and innovation-related projects very often falls heaviest on the household, which has virtually nothing to say about the timing and placing of

¹⁰This sub-project became routinized for many telegraph companies in the late 1850s, when line expansion was undertaken with the cooperation of railroads and offices were placed in railroad stations.

¹¹Although deserving lengthy treatment, this section does not discuss how the life paths of parents, and especially the status and income of their innovation-based employment roles, affect the household and neighborhood interactions and life content of their children.

activity bundles associated with employment, consumption, education, and institutions in general. Or, the relatively smooth functioning of a family, or household, generally requires the allocation of specialized one-person tasks that must be done by certain times and the coordination and juggling of paths so as to permit joint eating, playing, conversation, problem solving, and sexual activity; yet, very frequently, the outside-dictated daily routines and paths of different household members may fit together rather poorly for these purposes. (The formation of household activity bundles is likely to prove especially difficult when two or more household members have full-time employment roles.) In addition, the careful sequencing, synchronization, and synchORIZATION prescribed by innovation-based activities outside the household may lead to a high degree of stress for individual household members.

The combination of poorly matched daily programs and high stress levels may alter household-based life content in a multitude of negative ways (e.g., certain tasks may become unnecessarily male- or female-bound, thereby affecting sexual identity; playtime between parents and children, or between siblings, may become severely reduced or totally eliminated, thereby affecting childhood personality development; or mealtime talk, interpersonal problem solving, and other forms of conversation constantly may be marked by irritability or short-temperedness, thereby perhaps badly influencing how people outside the household are emotionally dealt with). In contrast, more temporally flexible and less stressful conditions stemming from use of a different set of technological and institutional innovations might contribute to some positive aspects of household-based life content. However, regardless of how well the daily routines of different household members are or are not in phase with one another, the taking up of additional innovation-derived outside roles by one person will almost certainly affect how and when other members form activity bundles within the household itself.

Although there is little documentary evidence concerning the household experiences of clerks, operators, messenger boys, and other telegraph-office employees, earlier comments on their lengthy working hours and daily paths allow us to surmise that they had extremely few workday opportunities for household interaction and activity-bundle formation except under conditions of exhaustion and stress.¹² Such limited household interactional possibilities doubtlessly often left a deep imprint on personality development and other aspects of life content. The situation normally must have been particularly difficult for clerks and messenger boys, who had much lower incomes than operators and office superintendents, since in all probability other members of their households also had to work at least eleven or more hours per day.

Linked Innovations

The impact of some technological and institutional innovations on the life content of interdependent human individuals is even more far-reaching than

¹²During the middle of the nineteenth century this was also true to varying degrees for European and U.S. manufacturing employees, as well as for large-city store clerks and delivery boys in the United States [25; 33, vol. 1].

hitherto indicated. The life-content impact of such innovations extends beyond the daily paths of its users and consumers and all those people affected by the daily paths of users and consumers, beyond the independently existing specialized roles either directly necessitated by its own operation or created by its own project-based multiplier effects. The attributes and enabling circumstances of the technological and institutional innovations in question are such that *they directly or indirectly facilitate the appearance of linked innovations that precipitate life-content impacts of their own*—influencing the daily paths and life paths of many people and households, affecting the population system-activity system matching process of one or more places, and bringing projects into being. Particularly in the early decades of its operation in the United States, when its network spread more rapidly and extensively than that of the railroad, the telegraph was an excellent example of such an innovation. It directly called forth major innovations in at least three activity areas by virtue of its attributes; and, through enabling capital to circulate, turn over, and accumulate more rapidly, it helped to facilitate countless other innovations, especially in the realm of manufacturing.

Until the telegraph brought cities into direct electronic contact with one another, virtually all of the nonlocal news contained in any U.S. newspaper had first appeared days or weeks before in the columns of other journals [40]. The highly ramified system of mutual journalistic plagiarism that prevailed was possible largely because newspaper editors were permitted to exchange copies of their products with one another free of charge via the mails. By the 1850s the telegraph was encouraging the breakdown of this system through the simultaneous dispatch of news stories to many cities before they had appeared in any newspaper and, more specifically, through the institutional innovation known as the Associated Press. This brought life-content consequences to a considerable number of people inasmuch as the functioning of the Associated Press itself required the creation of specialized independently existing roles, including that of the nonlocally based correspondent, and inasmuch as the daily project of producing a newspaper changed in nature, thereby also necessitating the birth (or redefinition) of specialized independently existing roles.

It has been said that, as practices initiated in the U.S. in 1852 by the New York and Erie Railroad became more widespread, the telegraph “revolutionized railway operation and management” [46]. Profit-motivated railroads that had telegraph lines paralleling their routes could innovate in the following ways: they could coordinate the dispatching of trains from their various stations, thereby reducing uncertainty about conditions ahead and permitting faster average operating speeds; they could make it generally known when trains were delayed between stations due to mechanical or other problems, thereby reducing the then extremely high frequency of accidents and on-line entanglements;¹³ they could expedite terminal operations by making it known in

¹³Contemporaries claimed that: “A single track railroad may be rendered more safe and efficient by a proper use of the telegraph than a double track railroad without its aid; as the double track can only obviate collisions that occur between trains moving in *opposite* directions, whereas the telegraph may be used effectually in preventing them either from trains moving in an opposite or the same direction” [1, vol. 33, 1860].

advance what the length and content of arriving freight trains was to be; and they could make more efficient use of rolling stock by allowing trains to be made up in accord with word received from ahead on expected demand for the day [1, 46]. The impacts on life content deriving from railroad innovations based on the telegraph were diverse. Passengers moving more quickly between origin and destination could insert additional activity bundles into their daily paths. The project of scheduling, dispatching, and operating trains between places was reshaped, thus fostering new and redefined specialized independently existing roles and affecting the life paths of their holders. Not least significantly, the paths of those who would have otherwise died in train accidents were extended.

Prior to the early 1850s fire-fighting services in large U.S. cities were limited in their ability to cope with major conflagrations. Alarms could not be spread rapidly, neighborhood units (usually consisting of volunteers) could not respond quickly, and assistance from beyond the neighborhood normally could be acquired only after considerable delay. Telegraph-using innovations made in Boston during 1851–52, and shortly thereafter in other major urban centers, radically modified fire-fighting services [10]. Through signal boxes that were telegraphically connected to local fire stations, alarms could be given much faster. In addition, numerous fire-fighting units could be assembled quickly at a single blaze, since the headquarters units of newly established city-wide fire departments were telegraphically linked up with all their subordinate stations (which were now grouped into divisions with specific areal jurisdictions). The life content impacts of these linked innovations were not confined to those previously holding fire-fighting roles and those now fitting into new specialized self-existing roles at fire-department headquarters. They also spread, in a very real sense, to those fire victims who experienced less damage than they might have under pre-telegraphic conditions.

The appearance of a very large family of economic innovations, each touching upon the daily and life paths of great numbers of people, was indirectly linked in part to the enabling circumstances provided by the telegraph. The telegraph enabled capital to circulate, turn over, and accumulate more rapidly than previously had been the case in the U.S. by:

1. allowing bankers, brokers, manufacturing entrepreneurs, and other businessmen in different cities to close deals, make sales, or extend credit more quickly, since they could now contact one another three or four times in a single day rather than consuming many days or weeks in making the same number of contacts via the mails;
2. permitting news of distant market changes to be more rapidly obtained and acted upon;
3. enabling retailers, wholesalers, and manufacturers to place distant orders on short notice, hence reducing their need to tie capital up in inventory; and
4. allowing input and output shipments to arrive more speedily as a consequence of the new faster average operating speeds of trains [1, 2, 25, 27, 43, 46].

The quicker capital could circulate, turn over, and accumulate, the more feasible it became for entrepreneurs either to expand the scale of their existing operations by investing in more fixed capital (*or technological innovations*) and physical inputs and possibly hiring more hands; or to invest in completely new manufacturing facilities (*or technological innovations*) that were large-scaled in nature, involved a finer division of labor (i.e., more specialized individually existing roles), and aided the fast turnover of capital by cutting down the time needed to produce a given quantity of goods.¹⁴ The scale-economy manufacturing at least partly facilitated by the telegraph's influence on capital accumulation could exploit the market-area expansion possibilities offered by the concurrently expanding railroad network. For this and other reasons (including agglomeration economies), such manufacturing tended to concentrate in large cities at the expense of much rural and small-town industrial employment [39]. In short, *through contributing to the adoption of a great variety of indirectly linked manufacturing innovations, the telegraph reached into the life content of hundreds of thousands of people by, among other things, helping to create or destroy production projects* (each with its own employment multipliers) *and significantly altering the population system-activity system matching process in most local areas.*

LIFE-CONTENT IMPACTS AND THE CONTRAPUNTAL DYNAMICS OF TIME-GEOGRAPHY

The thus far discussed life-content impacts of technological and institutional innovations can be restated and embellished upon relatively briefly in terms of the four contrapuntal, or antithetical, dynamics that are embedded in the time-geographic perspective on the human condition. (Alternatively, these contrapuntal, or antithetical, dynamics could be labeled "dialectics," but such a gesture would very probably cause confusion or require the raising of philosophical issues beyond the scope of this paper.) It is the simultaneous operation of these dynamics, as much as anything else, that makes the individual and collective impact of technological and institutional innovations on human individuals so extensive and deep reaching.

The individual-societal contrapuntal dynamic: There is nothing that time-geographically can affect a human individual without affecting society as a whole, nor is there anything that time-geographically can affect society as a whole without affecting a human individual. Any time a person freely or involuntarily commits a portion of his daily path and limited time resources to a technological or institutional innovation, he subtracts from the finite daily time resources of society as a whole in a given area, thereby diminishing the volume of other potential activities that can be incorporated into the time-space

¹⁴It is perhaps more than coincidental that in Marx's writings of the 1850s and 1860s [33, vol. 2; 34], when the economic impact of the telegraph—in combination with the railroad—was probably at its greatest in the United States and Great Britain, there is much preoccupation with the circulation and turnover times of capital and their relationship to capital accumulation and the development of large-factory based industrial capitalism.

organization of that areally defined segment of society. And, conversely, each innovationally increased (or decreased) claim made by the society-wide activity system on the daily (or biographical) time resources of a person reduces (or expands) the volume of other potential activity he is able to pack into his daily (or life) time-space path.

The daily path–life path contrapuntal dynamic: Nothing can touch a person's life path without influencing his daily path, and nothing can influence his daily path without touching upon his life path. When a person's life path becomes associated with (or disassociated from) a long-term independently existing role as the result of a technological or institutional innovation, he must add (or subtract) specific stations and activity bundles to his daily path. When the most immediately visible life-content impact of an innovation is at the daily-path level (e.g., the consumption of a newly offered good or service) it also has life-path implications for the person involved since, in an ultimate sense, the accumulation of one's daily paths is at the core of one's life path.

The external (corporeal action)–internal (mental experience and intention) contrapuntal dynamic: No human individual can wander over his physically observable daily and life paths, corporeally (or externally) participating in activity bundles and interactions, without accumulating mental (or internal) experiences that shape his attitudes, skills, and preferences, and hence his intentions; yet in choosing among activity bundles and interactions that are possible within the environment's time-geographic constraints, and eventually adding corporeal actions to his daily and life paths, no individual can escape the influence of his previous mental experiences and consequently derived intentions. Or, as a person moves from past to future along a constantly advancing now line, he is the center of a cumulative antithetical feedback between his corporeal actions and mental experiences and intentions. More specifically, when parts of a person's daily and life paths are corporeally channeled through time-space by the use or consumption of a technological or institutional innovation, he is exposed to environmental impulses, personal contacts, influences, and information in general that he would not otherwise mentally have experienced. Oppositely, when a person consciously or unconsciously employs the mental experiences acquired in connection with innovation use or consumption to formulate his intentions, and choose among alternatively possible actions and interactions, he sooner or later undertakes corporeal action that he would not have otherwise. (For important related remarks, see [4, 28, 50].)

The path convergence–path divergence contrapuntal dynamic: People do not live suspended in air. The time-space trajectory of every human individual is always proximate and contiguous to other people or objects, or in touch and together with the time-space trajectories of other people and nonhuman or inanimate "individuals." Therefore, no activity bundle or interaction can come into being via the convergence of paths without necessitating the divergence of previously joined paths; and no activity bundle can terminate, via the divergence of paths, without necessitating the convergence of paths in new combinations, or new activity bundles or interactions. That is, everything is an uninterrupted chain of time-space filling and abandonment, of creation and destruction, "linked together in a complex web" [24]. Or, "every situation is inevitably

rooted in past situations" [19] and inevitably becomes the root of subsequent situations.¹⁵ Thus, each activity bundle encompassed by the project(s) of a technological or institutional innovation becomes possible only upon the dissolution of all those immediately prior activity bundles and interactions involving the project bundle's human participants and resource inputs. Likewise, upon the completion of each activity bundle of an innovation use or consumption project, all the participating people (as well as any surviving objects and newly born inanimate outputs) are released to join, by choice or necessity, in immediately subsequent path combinations, or activities or interactions, the nature of which is constrained by the time-geographic characteristics of both the project bundle itself and the surrounding environment.

CONCLUDING REMARKS

Time-geographic realities, and especially the local connectedness of seemingly unrelated events and activities, insure that any technological or institutional innovation will have impacts on the life content of its users and consumers, as well as others, that are much more varied and profound than those identified by conventional forms of innovation assessment that normally focus almost entirely on myopic economic horizons. Regardless of personal concerns, a seeking of some understanding of these manifold impacts would appear to deserve a very high priority among scholars in highly industrialized societies, whether capitalistic or communistic; for it is in these societies where the greatest inroads into the life content of people already have been made by the centrally controlled projects and independently existing roles of technological and institutional innovations, and where such innovations will probably continue to amass most rapidly in the foreseeable future.

The need to cope on a day-to-day basis and the need to comprehend one's own immediate life and environment in a world where one is confronted by an often-confusing succession of technological and institutional innovations would also appear to be sufficient grounds for seeking some understanding of the life-content impacts of such innovations. Yet, with very few exceptions, the geographers of highly industrialized countries have been content to confine their study of technological and institutional innovations to questions pertaining either to the diffusion of isolated innovations (which are normally assumed to contribute to "development" and "progress" rather than mere change[8]), or to the spatial structures and organization generated by single innovations. Knowing where and how each class of props and backdrops has come to be placed on the enormous stage of the landscape may be important, and knowing how the different members of any particular class are related to one another in terms of location and attractive force exerted upon the players may also be important. But, it would seem that these things are nowhere near as important as knowing something about the way in which the total flow, timing, and sequencing of each new scene (or project) in the landscape drama is directed around both its own props and backdrops and previously existing props and backdrops: with

¹⁵This may also be true due to the external-internal contrapuntal dynamic.

every assignment of some action or personal contact to a player precluding or permitting other actions and contacts both on his part and the part of others in simultaneous and subsequent scenes (or projects); and with some preemptions of time-space causing old props or backdrops to be pushed aside.

A time-geographic approach to the study of existing or proposed technological and institutional innovations allows their actual or potential life-content impacts to be dealt with in both a humanistic and a deductive manner. It allows insights into activities and experiences to be gained in terms of integral human lives, rather than lives that have been fragmented into arbitrary and reified social-science categories. It allows dissimilar events and activities to be glimpsed in terms of their full societal complexity, in terms of collateral processes and the intricate local connectedness of people and things, rather than in terms of disciplinary vacuums created, and often stubbornly preserved, for analytical convenience [14, 22]. However, whatever a time-geographic perspective on technological and institutional innovations is capable of accomplishing, it can neither provide definitive answers nor address itself on its own to all the possible questions that might be raised concerning the life-content impacts of such innovations. For one thing, the full web of interdependent daily and life paths impacted by any innovation is too complicated to be dealt with in its entirety; although through the application of appropriate models and schema already employed by Hägerstrand and his followers, rather meager empirical data on activity timing and location will allow a lot to be said about the permitted and omitted path elements of either actual persons or representative hypothetical individuals. Furthermore, although the time-geographic approach can conceptualize the interrelationships between a person's accumulation of internal mental experiences and the external corporeal actions that constitute his daily and life paths, it cannot actually reveal the mental experiences, or internal life-content impacts, stemming from any technological or institutional innovation. Finally, some of the life-content impacts of technological and institutional innovations either cannot be addressed directly by time-geography (as in the case of traditionally assessed economic well-being impacts), or probably can be dealt with best through complementing time-geography with other relevant conceptual frameworks (as in the cases of alienation and the appearance of certain forms of collective protest [5, 37]).

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